

Letter Algebra: Formal and Evaluational Identities in the Alphabet of Powers

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Abstract

We treat the Latin letters as named monomials in $\mathbb{Z}[X]$, evaluated at an integer base $B \geq 2$. Concatenation of letters is addition of monomials, never multiplication. In this notation two classical facts become typographic. First, multiplying a sparse sum by a letter is a translation of the alphabet: $(Q + c) \cdot e = V + h$. Second, raising a scaled letter to a letter power peels the coefficient off as an ordinary integer in front of a monomial: $(2b)^a = 2^a \cdot b^a$, which at $B = 10$ is $1024 \cdot t$. We split identities into three layers—formal in $\mathbb{Z}[X]$, covariant after evaluation at every B , and named only at a particular base—and certify each layer by canonical polynomial equality, without expanding astronomical digit strings.

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1 Letters as monomials.

Fix an integer $B \geq 2$, the *calculator base*. Write X for an indeterminate. The *rank map* r sends $a, \dots, y$ to $1, \dots, 25$, $A, \dots, Y$ to $26, \dots, 50$, and Z (with alias z) to 100 .

The *letter embedding* is the map φ from letters to $\mathbb{Z}[X]$ given by $\varphi(\ell) = X^{r(\ell)}$. Evaluation $\text{ev}_B: \mathbb{Z}[X] \rightarrow \mathbb{Z}$ is $p \mapsto p(B)$. The number a letter *means* is $\text{ev}_B(\varphi(\ell)) = B^{r(\ell)}$. In base ten, $a = 10$, $b = 100$, $c = 10^3$, $e = 10^5$, $q = 10^{17}$, $t = 10^{20}$, $O = 10^{40}$, $Q = 10^{42}$, $h = 10^8$, $V = 10^{47}$, $Z = 10^{100}$.

A general value is a sparse polynomial $p = c_0 + \sum_i c_i X^{e_i}$ with $c_i \in \mathbb{Z}$ and $e_i \geq 1$. *Carry* (canonical form) rewrites any such p so that every coefficient lies in $\{0, 1, \dots, B-1\}$. After carry, ev_B is injective on canonical polynomials: this is ordinary uniqueness of base- B expansions [1].

Arithmetic is performed in $\mathbb{Z}[X]$. Decimal strings are requested only when a figure needs them. Equality of two expressions, written $L == R$, compares canonical polynomials of the same base. That is the certificate relation used below. It is not a comparison of decimal dumps.

2 Juxtaposition is addition.

Theorem 1 (Words are sparse sums). *A finite word in letters and digit-letter chunks denotes the sum of its chunks. In particular, for letters α, β ,*

$$\alpha\beta = \alpha + \beta = X^{r(\alpha)} + X^{r(\beta)}.$$

Juxtaposition never multiplies. Multiplication is the operator $$.*

Proof. This is the grammar of the language. The content is that the free commutative monoid on the alphabet, sent by φ , lands in the additive monoid of $\mathbb{Z}[X]$. $\square$

Certificate. $cQ == Q + c$ evaluates to 1. The word cQ is the two-spike polynomial $X^{42} + X^3$.

A glued block such as $2b$ or cQ is a single literal, so $2b^a$ is $(2b)^a$ and $cQ * e$ is $(c + Q) \cdot e$. That parsing is part of the notation, not an accident of precedence.

3 Translation of the alphabet.

Theorem 2 (Letter multiplication is rank addition). *For letters α, β ,*

$$\alpha * \beta = X^{r(\alpha)+r(\beta)}.$$

*If $r(\alpha) + r(\beta)$ is itself a named rank, the product is that letter. In base-independent form: $a * b = c$, $e * Q = V$, $e * c = h$.*

Proof. $\varphi(\alpha)\varphi(\beta) = X^{r(\alpha)}X^{r(\beta)} = X^{r(\alpha)+r(\beta)}$. $\square$

Theorem 3 (Translation). *Let $w = \sum_i \ell_i$ be a sum of letters and let ε be a letter. Then*

$$w * \varepsilon = \sum_i \ell'_i$$

where $r(\ell'_i) = r(\ell_i) + r(\varepsilon)$. In the running example,

$$(Q + c) \cdot e = V + h.$$

Proof. Distribute and apply Theorem 2: $r(Q) + r(e) = 42 + 5 = 47 = r(V)$ and $r(c) + r(e) = 3 + 5 = 8 = r(h)$. $\square$

Certificate. $cQ * e == V + h$ and $cQ * 2e == 2*(V+h)$.

The second identity is the first with a scaled letter: $2e = 2 \cdot X^5$. One side may be stored as the factored pair $2 \cdot (V + h)$ and the other as the distributed polynomial $2V + 2h$. Canonical equality identifies them (Theorem 7). A printer may still show $2 \cdot (V + h)$ on the left—that is the figure, not a disagreement.

This is Figure 1. In decimal form the same identity is a digit string that begins 100 and ends 01000 00000: the two spikes have moved five places. The letter line underneath *names* the move.

The rank map is a homomorphism from the multiplicative monoid of letters (under $*$) to $(\mathbb{N}, +)$. Translation by ε is the action of adding $r(\varepsilon)$ on ranks, hence a shift of the alphabet. The same action on a sparse sum is Theorem 3.

Expression	Polynomial	Named form
cQ	$X^{42} + X^3$	$Q + c$
$cQ * e$	$X^{47} + X^8$	$V + h$
$cQ * 2e$	$2X^{47} + 2X^8$	$2(V + h)$

Figure 1: Translation of a two-spike word by the letter e . Juxtaposition is the sum; $*$ shifts ranks.

4 Support gaps and carry.

Theorem 4 (Support gap). *Let $p, q \in \mathbb{Z}[X]$ be canonical, with $\text{supp}(p)$ and $\text{supp}(q)$ disjoint, and $\min \text{supp}(p) > \max \text{supp}(q)$. Then $p+q$ is already canonical, and the base- B digit string of $\text{ev}_B(p+q)$ is the digit string of $\text{ev}_B(p)$ followed by exactly $\min \text{supp}(p) - \text{width}(q)$ zeros, followed by the digit string of $\text{ev}_B(q)$. In particular there is no carry from q into p .*

Proof. Canonical form only carries when a coefficient is $\geq B$. Disjoint supports add coefficientwise with no collision, so no carry is created. The base- B layout of a polynomial is digits sitting at place-values B^e for e in the support; a gap in support is a run of zeros. $\square$

For $Q + c$ the gap is $42 - 3 = 39$ place-values. This is why cQ looks like a 100 and a 01000 with nothing in between, and why adding c to Q does not perturb the high spike.

Carry *does* matter for identities such as $10 == a$ in base ten: the constant 10 is rewritten as X^1 . Formal division in $\mathbb{Z}[X]$ and integer division after carry can therefore disagree about the *route*, even when they agree about the integer quotient. That distinction belongs in a remark, not the lead.

5 The coefficient-power law.

Theorem 5 (Coefficient-power). *Let $k \in \mathbb{Z}$ and let $n, m \geq 0$. After evaluation at B ,*

$$(k \cdot B^n)^{B^m} = k^{B^m} \cdot B^{nB^m}.$$

In letters: $(kb)^{B^m}$ with $r(b) = n$ is k^{B^m} times the monomial of rank nB^m . Taking $k = 2$, $n = 2$, $m = 1$:

$$(2b)^a = 2^a \cdot b^a.$$

Proof. $(kB^n)^{B^m} = k^{B^m} (B^n)^{B^m} = k^{B^m} B^{nB^m}$. $\square$

This identity is *not* an identity in $\mathbb{Z}[X]$. The exponent a is the integer B , not the indeterminate. It is nevertheless true for every $B \geq 2$. Call such identities *base-covariant*.

At $B = 10$, $2^a = 2^{10} = 1024$ and $b^a = (10^2)^{10} = 10^{20} = t$, so $(2b)^a = 1024 \cdot t$. Doubling the exponent, $(2b)^{2a} = 2^{20} \cdot b^{2a} = 1048576 \cdot O$. Binary powers appear as coefficients of decimal monomials. That is Figure 2.

A value may be stored as a pair (k, p) : a leading coefficient times a sparse polynomial. Raising the scaled monomial $2b = 2 \cdot X^2$ to the tenth power keeps the factor, printed $1024 \cdot t$. Multiplying the already-carried integer 2^a by t distributes that factor and carries, printed $w + 2u + 4t$. Both evaluate to the same twenty-four digit integer 1024 followed by twenty zeros. They are the same integer in two presentations.

Lemma (Digit expansion of a scaled monomial). *For $k \geq 0$ and $e \geq 0$, if $k = \sum_{i=0}^d k_i B^i$ is the canonical base- B expansion of k , then*

$$k \cdot X^e = \sum_{i=0}^d k_i X^{e+i}$$

in $\mathbb{Z}[X]/(X - B)$ after carry. At $B = 10$, $k = 1024$, $e = 20$, this is $1024 \cdot t = w + 2u + 4t$.

Proof. Carry is the rewriting $B^i X^e = X^{e+i}$ in the quotient by $X - B$. Apply it to each digit of k . $\square$

Certificate. $(2b)^a == 2^a * (b^a)$ and $2b^a == 1024 * t$ both evaluate to 1. The printed strings remain distinct: that is the lemma, not a failure of the identity. Subtraction of two presentations is not the certificate; $==$ is.

Expression	Factored form	Carried form
b^a	t	t
$(2b)^a$	$1024 \cdot t$	$w + 2u + 4t$
$(2b)^{2a}$	$1048576 \cdot O$	carry of $2^{20} X^{40}$

Figure 2: Coefficient-power at $B = 10$. The factor 2^{B^m} sits in front of a decimal monomial; carry redistributes it as digits.

6 Formal versus evaluational identities.

Definition 1. An identity $L = R$ written in letters is

- *Layer I* (formal) if it holds in $\mathbb{Z}[X]$ before evaluation (Theorems 1–3);
- *Layer II* (covariant) if $\text{ev}_B(L) = \text{ev}_B(R)$ for every $B \geq 2$, while the two sides need not be the same element of $\mathbb{Z}[X]$ before the exponent is evaluated (Theorem 5)—after canonical presentation they still compare equal;

- *Layer III* (named) if it uses a letter or a decimal numeral on the right-hand side whose rank (or value) equals a B -dependent quantity, so it holds only for particular B (Theorem 6).

The same identities, re-evaluated at a new base, separate the layers. Changing the base does not convert a finished decimal dump; it reinterprets every letter as a power of the new B .

Theorem 6 (Naming at base ten). *The following are equivalent to $B = 10$:*

1. $b^a = t$, because $b^a = B^{2B}$ and $t = B^{20}$;
2. $b^{2a} = O$, because $2B \cdot B = 40$;
3. $2^a = 1024$.

Proof. Immediate from the rank map and Theorem 5. □

At every base the Layer I and Layer II certificates remain 1. The Layer III certificates become 0. Table 1 records the experiment at $B = 2, 10$, and 16. Figure 3 shows where the names went.

Identity	Layer	$B = 2$	$B = 10$	$B = 16$
$cQ * e == V + h$	I	1	1	1
$(2b)^a == 2^a * (b^a)$	II	1	1	1
$b^a == t$	III	0	1	0
$2^a == 1024$	III	0	1	0
$2b^a == 1024 * t$	III	0	1	0

Table 1: The same certificates at three bases. Layers I and II are invariant; named equalities live at $B = 10$.

Expression	$B = 2$	$B = 10$	$B = 16$
a	2, a	10, a	16, a
b^a	16, d	10^{20} , t	16^{32} , G
t	2^{20} , t	10^{20} , t	16^{20} , t
2^a	4, b	1024, $c + 2a + 4$	65536, d
$(2b)^a$	64, $4 \cdot d$	$1024 \cdot 10^{20}$, $1024 \cdot t$	$65536 \cdot 16^{32}$, $65536 \cdot G$
$2^a \cdot b^a$	64, f	same integer, $w + 2u + 4t$	same integer, K

Figure 3: Values, not certificates. Each cell is the integer and its printed monomial form. Letters name ranks, not decimal numbers.

Letters name *ranks*, not decimal numbers. At $B = 2$, $b^a = (2^2)^2 = 2^4 = d$, while t is still the letter of rank 20. The identity $b^a = t$ asked a rank-20 letter to equal a rank that depends on B . It fails as soon as $2B \neq 20$. The numeral 1024 is the same kind of object: it is 2^{10} , hence Layer III.

Layer II never mentions those names. $(2b)^a = 2^a \cdot b^a$ holds at $B = 2$ as $64 = 64$, and at $B = 16$ as two forty-digit integers that `==` identifies without printing them.

At $B = 2$ the factored form of $(2b)^a$ is $4 \cdot d$ and the carried form is f , because $4 \cdot X^4 = X^6$ after carry in base 2. At $B = 16$ the coefficient $2^{16} = 65536 = 16^4$ is itself a monomial, so $65536 \cdot G$ carries all the way to the single letter K (rank $32 + 4 = 36$). At $B = 10$ the coefficient 1024 is *not* a power of ten, so the factored line $1024 \cdot t$ and the carried line $w + 2u + 4t$ remain visibly different. Canonical equality identifies all three pairs (Theorem 7). The printer is allowed to show the factor; the certificate is not allowed to care.

Remark (When 2^B is a monomial). $2^B = B^k$ for an integer $k \geq 0$ if and only if $B = 2^{2^m}$ for some $m \geq 0$ ($B = 2, 4, 16, 256, \dots$). In that case 2^a is a unit monomial and the two printed presentations of $(2b)^a$ collapse to one letter after carry. Base ten is interesting precisely because it is *not* on that list: the binary coefficient is forced to sit in front of a decimal monomial, which is the picture in Figure 2.

Proof. If B has an odd prime factor then B^k does too, while 2^B does not. Write $B = 2^s$. Then $2^{sk} = 2^B$ gives $k = 2^s/s$, so $s \mid 2^s$, hence s is a power of 2. $\square$

7 Canonical presentation.

An Alphabet of Powers value is a pair $(k, p) \in \mathbb{Z} \times \mathbb{Z}[X]$. The empty polynomial is the unit for multiplication by k : it denotes the constant k , equivalently kX^0 . Write $D(k, p) = k \cdot p \in \mathbb{Z}[X]$ for the *distributed* polynomial, and write C for carry into digits in $\{0, 1, \dots, B-1\}$.

Theorem 7 (Canonical presentation). *For non-negative values, $C(D(k, p))$ is the unique canonical base- B expansion of the integer $k \cdot p(B)$. In particular*

$$(k, p) \sim (k', p') \quad \text{iff} \quad C(D(k, p)) = C(D(k', p'))$$

if and only if $\text{ev}_B(kp) = \text{ev}_B(k'p')$. The relation $\sim$ does not require writing that integer.

Proof. Uniqueness of base- B expansions of non-negative integers. Carry is exactly the rewriting $cX^e \mapsto (c \bmod B)X^e + \lfloor c/B \rfloor X^{e+1}$ until every coefficient is a digit. Two polynomials with digits in $\{0, \dots, B-1\}$ evaluate equally if and only if they are identical. $\square$

The relation `==` is $\sim$. The stored pair (k, p) may be printed without forcing $C \circ D$, which is why $(2b)^a$ still reads $1024 \cdot t$ while $2^a \cdot b^a$ reads $w + 2u + 4t$. The certificate does not care. Decimal evaluation is ev_B , used for figures, not for equality.

A glued comparison such as `cQ == Q + c` requires `==` to bind looser than `+`. Without that, the parser would read `(cQ == Q) + c` and return the letter c .

8 Closing remarks.

We do not claim a new theorem of number theory. We claim a language in which two classical identities are *seen*, a three-layer split that base change makes experimental, and a certificate relation that does not require writing the numbers the identities are about.

Hyperpower collapse— $Z^e = a^g$ at base ten, without writing 10^{10^7} digits—is a companion note. The multinomial theorem, and the identity $\gcd(B^i, B^j) = B^{\min(i,j)}$ on letters, remain further sequels.

References

- [1] D. E. Knuth, *The Art of Computer Programming*, Vol. 2: *Seminumerical Algorithms*, 3rd ed., Addison-Wesley, Reading, MA, 1997.